

Question ID d72a2b4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d72a2b4d

In March, the price of a collectible card was **\$15.50**. In April, the price of the collectible card was **\$17.36**. The price of the collectible card in April was $p\%$ of the price of the collectible card in March. What is the value of p ?

- A. 12
- B. 88
- C. 112
- D. 188

ID: d72a2b4d Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that the price of the collectible card was **\$15.50** in March and **\$17.36** in April. It's also given that the price of the collectible card in April was $p\%$ of the price in March. It follows that **\$17.36** is $p\%$ of **\$15.50**. Therefore, the value of p can be calculated by solving the equation $17.36 = \left(\frac{p}{100}\right)(15.50)$, or $17.36 = \frac{15.50p}{100}$. Multiplying each side of this equation by **100** yields $1,736 = 15.50p$. Dividing each side of this equation by **15.50** yields $112 = p$. Therefore, the value of p is **112**.

Choice A is incorrect. **12%** is the percent increase in the price of the collectible card from March to April.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID ad1d6adb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: ad1d6adb

The number of coins in a collection increased from **9** to **90**. What was the percent increase in the number of coins in this collection?

- A. 10%
- B. 81%
- C. 90%
- D. 900%

ID: ad1d6adb Answer

Correct Answer:
D

Rationale
Choice D is correct. It's given that the number of coins in the collection increased from **9** to **90**. It follows that the number of coins in the collection increased by $90 - 9$, or **81**. Let $x\%$ represent the percentage that **81** is of **9**. The value of x can be found using the proportion $\frac{81}{9} = \frac{x}{100}$, or $9 = \frac{x}{100}$. Multiplying both sides of this equation by **100** yields $900 = x$. Thus, when the number of coins in the collection increased from **9** to **90**, the percent increase was **900%**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 566759ef

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 566759ef

Thomas installed a new stove in his restaurant. At the time of installation, the stove had a value of \$800. Thomas estimates that each year the value of the stove will depreciate by 20% of the previous year’s estimated value. What is the estimated value of the stove exactly 2 years after Thomas installed it?

- A. \$480
- B. \$512
- C. \$556
- D. \$640

ID: 566759ef Answer

Rationale

Choice B is correct. If the stove’s value depreciates by 20% of the previous year’s estimated value, then each year it retains $100\% - 20\% = 80\%$, or 0.80, of the previous year’s estimated value. Since the stove’s value was \$800 when Thomas installed it, the estimated value after two years would be $(0.80)(0.80)(\$800) = \512 .

Choice A is incorrect. This is the value of the stove if each year it had depreciated by 20% of the original value rather than by 20% of the previous year’s estimated value. Choice C is incorrect and may be the result of a computational error. Choice D is incorrect. This is the estimated value of the stove 1 year after Thomas installed it, not 2 years.

Question Difficulty:
Medium

Question ID dccdb20c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: dccdb20c

There are **450** tiles in a box. Of these tiles, **6%** are black. How many black tiles are in the box?

ID: dccdb20c Answer

Correct Answer:

27

Rationale

The correct answer is **27**. It's given that **6%** of the **450** tiles in a box are black. Therefore, the number of black tiles in the box can be calculated by multiplying the number of tiles in the box by $\frac{6}{100}$, which is equivalent to **450** $\left(\frac{6}{100}\right)$, or **27**.

Question Difficulty:

Medium

Question ID e9f4521a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: e9f4521a

13 is $p\%$ of 25. What is the value of p ?

ID: e9f4521a Answer

Correct Answer:

52

Rationale

The correct answer is 52. It's given that 13 is $p\%$ of 25. It follows that $\frac{13}{25} = \frac{p}{100}$. Multiplying both sides of this equation by 100 gives $52 = p$. Therefore, the value of p is 52.

Question Difficulty:

Medium

Question ID 7b731fc3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 7b731fc3

What number is 40% greater than 115?

ID: 7b731fc3 Answer

Correct Answer:

161

Rationale

The correct answer is 161. For a number to be 40% greater than 115, it follows that the number is (100% of 115) + (40% of 115), which can be written as $\frac{100}{100}(115) + \frac{40}{100}(115)$. This expression is equivalent to $1(115) + 0.4(115)$, or $1.4(115)$, which is equal to 161. Therefore, 161 is 40% greater than 115.

Question Difficulty:

Medium

Question ID 94c65646

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 94c65646

432 is 96% of what number?

ID: 94c65646 Answer

Correct Answer:
450

Rationale

The correct answer is 450. Let x represent the number that 432 is 96% of. This can be written as $(\frac{96}{100})x = 432$, or $0.96x = 432$. Dividing both sides of this equation by 0.96 yields $x = 450$. Therefore, 432 is 96% of 450.

Question Difficulty:
Medium

Question ID 7f84b136

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 7f84b136

The table summarizes the number of objects in each group.

Group	Number of objects
A	375
B	54
C	690
D	81
Total	1,200

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p ?

ID: 7f84b136 Answer

Correct Answer:
184

Rationale

The correct answer is **184**. It's given in the table that there are **375** objects in group A and **690** objects in group C. It's also given that the number of objects in group C is $p\%$ of the number of objects in group A. Therefore, **690** is $p\%$ of **375**, which can be represented by $690 = \left(\frac{p}{100}\right)(375)$, or $690 = 3.75p$. Dividing both sides of this equation by **3.75** yields $184 = p$. Therefore, the value of p is **184**.

Question Difficulty:
Medium

Question ID c256b723

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: c256b723

The amount of Hanna's bill for a food order was **\$50**. Hanna gave a tip of **20%** of the amount of the bill. What is the amount, in dollars, of the tip Hanna gave?

ID: c256b723 Answer

Correct Answer:
10

Rationale

The correct answer is **10**. It's given that the amount of Hanna's food order was **\$50** and that Hanna gave a tip of **20%** of the amount of the bill. **20%** of **50** can be calculated as $\left(\frac{20}{100}\right)(50)$, which yields $\frac{1000}{100}$, or **10**. Therefore, the amount, in dollars, of the tip Hanna gave is **10**.

Question Difficulty:
Medium

Question ID b2f6f17d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: b2f6f17d

A customer’s monthly water bill was \$75.74. Due to a rate increase, her monthly bill is now \$79.86. To the nearest tenth of a percent, by what percent did the amount of the customer’s water bill increase?

- A. 4.1%
- B. 5.1%
- C. 5.2%
- D. 5.4%

ID: b2f6f17d Answer

Correct Answer:
D

Rationale

Choice D is correct. To find the percent increase of the customer’s water bill, the absolute increase of the bill, in dollars, is divided by the original amount of the bill, and the result is multiplied by 100%, as follows: $\frac{79.86 - 75.74}{75.74} \approx 0.054$; $0.054 \times 100\% = 5.4\%$.

Choice A is incorrect. This choice is the difference $79.86 - 75.74$ rounded to the nearest tenth, which is the (absolute) increase of the bill’s amount, not its percent increase. Choice B is incorrect and may be the result of some calculation errors. Choice C is incorrect and is the result of dividing the difference between the two bill amounts by the new bill amount instead of the original bill amount.

Question Difficulty:
Medium

Question ID 96a45430

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 96a45430

A number n is increased 6%. If the result is 318, what is the value of n ?

- A. 199
- B. 299
- C. 300
- D. 337

ID: 96a45430 Answer

Correct Answer:
C

Rationale

Choice C is correct. The decimal equivalent of 6% is 0.06. Since increasing the number n by 6% yields the number 318, this situation can be represented by the equation $n(1 + 0.06) = 318$, or $n(1.06) = 318$. Dividing both sides of this equation by 1.06 yields $n = 300$.

Choice A is incorrect. This is the result when n is increased by 60%, not by 6%. Choice B is incorrect. This is the approximate result of decreasing 318 by 6%. Choice D is incorrect. This is the approximate result of increasing 318 by 6%.

Question Difficulty:
Medium

Question ID 8cbf1415

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 8cbf1415

In a group, **40%** of the items are red. Of all the red items in the group, **30%** also have stripes. What percentage of the items in the group are red with stripes?

- A. **10%**
- B. **12%**
- C. **70%**
- D. **75%**

ID: 8cbf1415 Answer

Correct Answer:
B

Rationale

Choice B is correct. It’s given that in a group, **40%** of the items are red. It follows that the number of red items in the group can be represented by **0.4x**, where **x** represents the total number of items in the group. It’s also given that of all the red items in the group, **30%** also have stripes. It follows that the number of items in the group that are red and have stripes can be represented by **0.3(0.4x)**, or **0.12x**. The expression **0.12x** represents **12%** of **x**. Since **x** represents the total number of items in the group, it follows that **12%** of the items in the group are red and have stripes.

Choice A is incorrect and may result from subtracting **30%** from **40%** rather than calculating **30%** of **40%**.

Choice C is incorrect and may result from adding **30%** and **40%** rather than calculating **30%** of **40%**.

Choice D is incorrect and may result from calculating the percentage that **30%** is of **40%** rather than calculating **30%** of **40%**.

Question Difficulty:
Medium

Question ID ba61d95f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: ba61d95f

The population of Greenville increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7

ID: ba61d95f Answer

Correct Answer:
C

Rationale

Choice C is correct. Let x be the 2015 population of Greenville. It's given that the population increased by 7% from 2015 to 2016. The increase in population can be written as $(0.07)x$. The 2016 population of Greenville is given as the sum of the 2015 population of Greenville and the increase in population from 2015 to 2016. This can be rewritten as $x + (0.07)x$, or $1.07x$. Therefore, the value of k is 1.07.

Choice A is incorrect. This is the percent, represented as a decimal, that the population increased from 2015 to 2016, not the value of k .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

Question Difficulty:
Medium

Question ID 709e04de

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 709e04de

The value of z is **1.13** times **100**. The value of z is what percent greater than **100**?

- A. **11.3**
- B. **13**
- C. **130**
- D. **213**

ID: 709e04de Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that the value of z is **1.13** times **100**. This can be written as $z = (1.13)(100)$, which is equivalent to $z = (1 + 0.13)(100)$, or $z = (1 + \frac{13}{100})(100)$. It follows that the value of z is **100%** of **100** plus **13%** of **100**. Therefore, the value of z is **13%** greater than **100**.

Choice A is incorrect. This gives a value of z that is **1.113**, not **1.13**, times **100**.

Choice C is incorrect. This gives a value of z that is **2.30**, not **1.13**, times **100**.

Choice D is incorrect. This gives a value of z that is **3.13**, not **1.13**, times **100**.

Question Difficulty:

Medium

Question ID 8e2e424e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 8e2e424e

The number k is 36% greater than 50. If k is the product of 50 and r , what is the value of r ?

- A. 36
- B. 3.6
- C. 1.36
- D. 0.36

ID: 8e2e424e Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that the number k is 36% greater than 50. Therefore, the value of k is the number 50 plus 36% of 50.

This can be rewritten as $k = 50 + \left(\frac{36}{100}\right)(50)$. Multiplying the terms $\left(\frac{36}{100}\right)(50)$ yields 18, so $k = 50 + 18$, or $k = 68$. It’s also given that k is the product of 50 and r , which can be rewritten as $k = 50r$. Substituting 68 for k yields $68 = 50r$. Dividing both sides of this equation by 50 yields $r = 1.36$.

Choice A is incorrect. This is the percentage that k is greater than 50. Choice B is incorrect and may result from a calculation error. Choice D is incorrect. This would be the value of r if k were 36% of 50, instead of 36% greater than 50.

Question Difficulty:
Medium

Question ID 8a714fa1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 8a714fa1

Which of the following represents the result of increasing the quantity x by 9%, where $x > 0$?

- A. $1.09x$
- B. $0.09x$
- C. $x + 9$
- D. $x + 0.09$

ID: 8a714fa1 Answer

Correct Answer:
A

Rationale

Choice A is correct. Increasing the positive quantity x by 9% is the result of adding 9% of x to x . 9% of x can be represented algebraically as $\frac{9}{100}x$, or $0.09x$. Adding this expression to x yields $x + 0.09x$, or $1.09x$.

Choice B is incorrect. This represents 9% of x . Choice C is incorrect. This represents increasing x by 9, not by 9%. Choice D is incorrect. This represents increasing x by 0.09, not by 9%.

Question Difficulty:
Medium

Question ID 121dc44f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 121dc44f

The population of City A increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

- A. 0.07
- B. 0.7
- C. 1.07
- D. 1.7

ID: 121dc44f Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the population of City A increased by 7% from 2015 to 2016. Therefore, the population of City A in 2016 includes 100% of the population of City A in 2015 plus an additional 7% of the population of City A in 2015. This means that the population of City A in 2016 is 107% of the population in 2015. Thus, the population of City A in 2016 is $\frac{107}{100}$, or 1.07, times the 2015 population. Therefore, the value of k is 1.07.

Choice A is incorrect. This would be the value of k if the population in 2016 was 7% of the population in 2015.

Choice B is incorrect. This would be the value of k if the population in 2016 was 70% of the population in 2015.

Choice D is incorrect. This would be the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

Question Difficulty:
Medium

Question ID 7cab9fe1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 7cab9fe1

Which expression represents the result of increasing a positive quantity w by **43%**?

- A. **1.43** w
- B. **0.57** w
- C. **43** w
- D. **0.43** w

ID: 7cab9fe1 Answer

Correct Answer:

A

Rationale

Choice A is correct. The result of increasing a positive quantity w by $x\%$ can be represented by the expression $(1 + \frac{x}{100})w$. Therefore, the result of increasing a positive quantity w by **43%** can be found by substituting **43** for x in the expression $(1 + \frac{x}{100})w$, which gives $(1 + \frac{43}{100})w$, or **1.43** w . Thus, the expression **1.43** w represents the result of increasing a positive quantity w by **43%**.

Choice B is incorrect. This is the result of decreasing a positive quantity w by **43%**.

Choice C is incorrect. This is the result of increasing a positive quantity w by **4,200%**.

Choice D is incorrect. This is the result of decreasing a positive quantity w by **57%**.

Question Difficulty:

Medium

Question ID d693f563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d693f563

Last year, Cedric had **35** plants in his garden. This year, the number of plants in Cedric’s garden is **60%** greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

ID: d693f563 Answer

Correct Answer:
56

Rationale

The correct answer is **56**. It’s given that Cedric had **35** plants in his garden last year and that the number of plants in Cedric's garden this year is **60%** greater than the number of plants in his garden last year. It follows that the number of plants in Cedric’s garden this year is **35** plus **60%** of **35**, which is equal to **$35 + 35\left(\frac{60}{100}\right)$** , or **$35 + 35(0.6)$** . This expression is equivalent to **$35 + 21$** , or **56**. Therefore, Cedric has **56** plants in his garden this year.

Question Difficulty:
Medium

Question ID 63573fea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 63573fea

During the first month of sales, a company sold 1,300,000 units of a certain type of smartphone. During the same month, 15% of the units sold were returned. If sales and the return rate remain the same for each of the next 5 months, about how many units of this smartphone will be returned to the company during this 6-month period?

- A. 195,000
- B. 975,000
- C. 1,170,000
- D. 6,630,000

ID: 63573fea Answer

Correct Answer:
C

Rationale

Choice C is correct. Of the 1,300,000 units sold during the first month, 15% were returned, so $(1,300,000)(0.15) = 195,000$ units were returned during the first month. If the units were sold and returned at the same rate for the next 5 months, then a total of $(195,000)(6) = 1,170,000$ smartphone units were returned during the 6-month period.

Choice A is incorrect. This is the number of units that were returned in 1 month. Choice B is incorrect. This is the number of units that were returned in 5 months. Choice D is incorrect. This is the number of units sold and not returned during the first 6 months.

Question Difficulty:
Medium

Question ID 022e9894

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 022e9894

An insurance company offers a series of three information sessions. **1,250** people attended the first information session. **72%** of the people who attended the first information session attended the second information session, and **36%** of the people who attended the first and second information sessions attended the third information session. How many people attended all three information sessions?

ID: 022e9894 Answer

Correct Answer:

324

Rationale

The correct answer is **324**. It's given that **1,250** people attended the first information session, and that **72%** of the people who attended the first information session attended the second information session. Therefore, the number of people who attended the first and second information sessions can be found by calculating **72%** of **1,250**, which is equal to $1,250\left(\frac{72}{100}\right)$, or **900**. It's also given that **36%** of the people who attended the first and second information sessions attended the third information session. Since **900** people attended the first and second information sessions, the number of people who attended the first, second, and third information sessions can be found by calculating **36%** of **900**, which is equal to $900\left(\frac{36}{100}\right)$, or **324**. Therefore, **324** people attended all three information sessions.

Question Difficulty:

Medium

Question ID 05195d8a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 05195d8a

In a sample, 80% of the items are faulty. There are 88 faulty items in the sample. How many total items are in the sample?

ID: 05195d8a Answer

Correct Answer:

110

Rationale

The correct answer is 110. Let x represent the total number of items in the sample. It's given that 80% of the items are faulty and that there are 88 faulty items in the sample. Therefore, 80% of x is 88. Since 80% can be rewritten as $\frac{80}{100}$, it follows that $\frac{80}{100}x = 88$. Multiplying both sides of this equation by 100 yields $80x = 8,800$. Dividing both sides of this equation by 80 yields $x = 110$. Therefore, there are 110 total items in the sample.

Question Difficulty:

Medium